

IMPACT OF DYNAMIC LINE RATING ON THE GERMAN ELECTRICAL ENERGY SYSTEM

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ABSTRACT

With the deployment of renewable energy sources, grid capacity expansion is essential to transmit the energy where it is needed. Especially, when wind energy is generated in the north of Germany and then transmitted to the industrial south [1]. However, in Germany several grid capacity expansion projects have been delayed due to administrative problems and protest activities [2]. The longer commissioning time of new distribution lines compared to wind turbines further intensifies the problem of lacking transmission capacity [3].

The transmission capacity directly depends on the ampacity of the overhead transmission lines (OTL). Traditionally, OTL ampacity is calculated at unfavorable, static weather conditions [3]. In literature, the static calculation is referred to as static line rating (SLR). SLR underestimates the ampacity of OTL in energy models and leads to an overestimated grid capacity. In contrast to SLR, dynamic line rating (DLR) considers time varying weather conditions to calculate OTL ampacity. Therefore, the implementation of DLR has several benefits such as economic benefits, congestion reduction and better wind energy integration [4].

In literature, case studies consider DLR in simulations [3] or in small scale energy system optimization models (ESOM) [5] [6] [7]. However, no case study evaluates the impact of DLR in a higher scale ESOM e.g. national ESOM.

In this paper, we analyze the impact of DLR on the German electrical energy system (EES). For this, we build an ESOM in the PyPSA-Eur framework [8]. The model is hourly resolved and considers a time horizon of one year. To calculate DLR in our model, we use the IEEE standard [9] with highly resolved historical weather data using atlite [10]. With our model, we compare the German EES using SLR and DLR. For the comparison, we analyze wind curtailment, the grid congestion as well as the generator deployment of both energy systems. This allows us to illustrate the benefits of DLR in the German EES.

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